Sampling Frame for the rCARI Validation Study

rCARI study team

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1 Background

This document outlines the sampling methodology for the rCARI validation study with minor adjustments for country specific studies. The first section presents the study's generic methodology, while the subsequent sections present an adaption of the main methodology to reflect the context of particular pilot study country.

The study has been designed and implemented as a survey experiment with stratified sample. The study designed has been designed with the following considerations.

- Phone ownership & accessibility within the pilot country
- Distribution of food insecurity
- Areas of accessibility within the pilot countries
- Project budget

2 Sampling design & frame

Figure 1 below presents the generic sampling procedure that has been implemented in the 3 pilot countries so far. As mention earlier, selection and inclusion of study area followed a 3-point criterium: i.e., 1) phone coverage, 2) accessibility, and 3) food security situation. Phone coverage is important consideration for the remote interviews, while field accessibility ensures that enumerators can successfully access and interview households in face-to-face. The food security consideration would ensure that at least areas that have different AFI (acute food insecurity) situations are adequately represented in the study sample.

In all 3 pilot countries thus far, the same method was applied in identifying and selecting all Admin areas that have been included in the study.

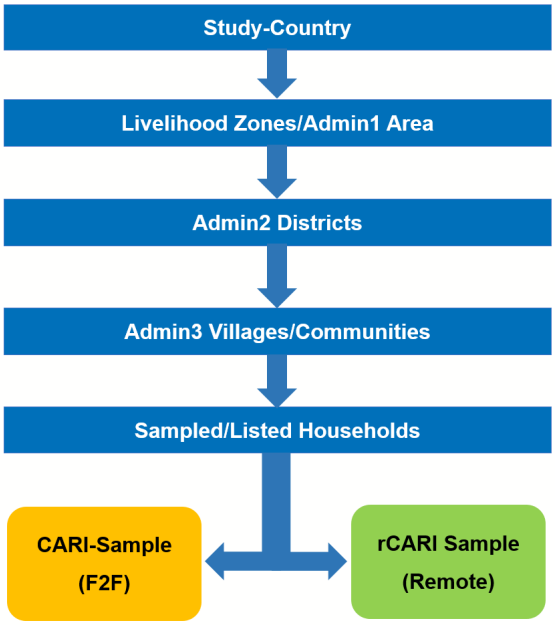

Figure 1: Sampling and Randomization Procedure

The main steps followed in identifying and selecting the admin areas at the various stages are listed and explained below.

Step 1 - Livelihood Zones/ADMIN1 Selection

Livelihood Zones/ADMIN1 Area Selection: A minimum of 2 Admin1 Areas to be identified together with WFP country offices (CO) based on phone coverage, food security situation and accessibility. In countries where more than 2 Admin1 areas meets the inclusion criteria, a simple random approach shall be followed to select the 2 Areas as study locations.

Step 2 - ADMIN2 Selection

In the second step, a minimum of 4 ADMIN2 areas were randomly selected (2 per ADMIN1 Area based on the outcome of Step 1) from the ADMIN1 areas that have been identified and selected.

Step 3 - ADMIN5 (villages, communities, or camps) Selection

All ADMIN 5 (villages) in ADMIN2 that meet the inclusion criteria set out in Step 1 are now compiled at this stage as the primary sampling units (PSUs). A minimum of 20 PSUs were randomly selected (distributed uniformly across the selected ADMIN1 AREAS) as sampling area to draw the study households from.

Step 4 - Sampled households

The sampling frame for the study was all households in selected PSUs. A minimum of 100 households per PSU were selected through the random-walk approach (listing every 3rd house in the sequence) by trained enumerators. Consent of participation was taken from all HHs before being included in the study sample. Based on the number of PSUs per country, 2000-2500HHs (except in Myanmar where the initial sample was 1,665HHs). The final listed households in this stage were used for the randomization process in the next step.

Step 5 - Randomization of household (hh)

Consented final sampled households in Step 4 are randomly assigned into one of two survey modalities/groups; F2F (CARI group) and Remote (rCARI group). The design is such that the sample will be divided equally between the modalities. All interviews were conducted over the same time frame to ensure that households in the two modalities are reporting their food security situation over the same window.

2.1 Myanmar

In Myanmar, the study was implemented amongst IDPs (internally displaced persons) in two regions: Kachin and Shan (Admin 1). The sampling list was compiled using WFP beneficiaries made up of IDPs in the two regions. A total of 20 camps were selected randomly as the primary sampling units (PSU). All households in the selected camps formed the sampling frame for the study in Myanmar. A minimum of 100 households were then randomly selected from each camp for the study (However, because some of the camps had fewer than 100 households, the final sampled households were 1665 in total).

2.2 Ecuador

In Ecuador, the study was implemented in three regions: Amazonia, Coasta and Sierra (Admin1 areas). In addition to the standard selection criteria based on accessibility and food security prevalence, consideration was also given to the safety of the team, considering the political situation at the time of implementing the study.

A total of 27 Admin5 areas (villages) were randomly selected from the 3 Admin1 areas based on the foregoing criteria. These formed the PSUs for the study in Ecuador. All households in the selected 27 PSUs formed the selection frame and were eligible to be enlisted into the study. A total of 3220 households were listed (approximately 120 households per PSU were randomly selected) for the study in two phases.

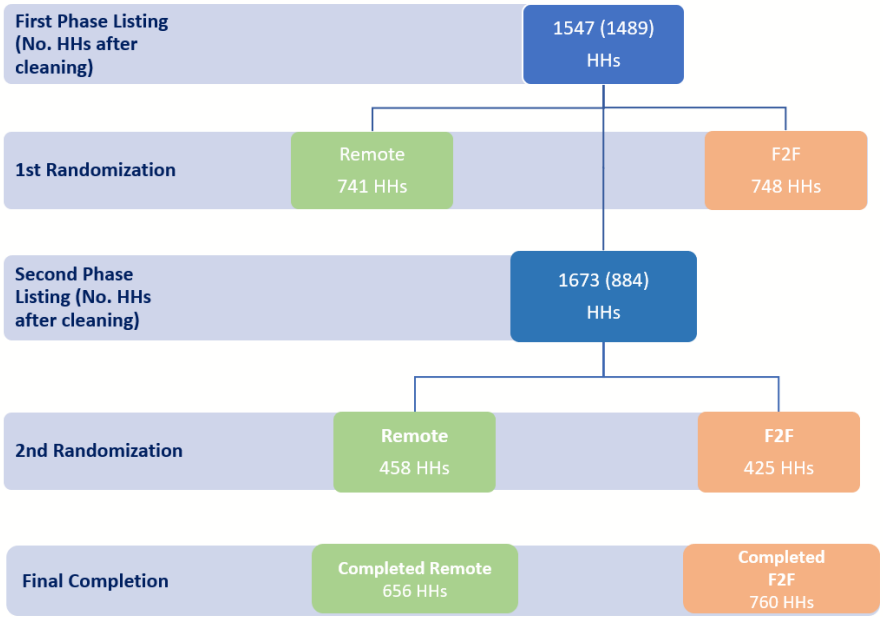

Figure 2: Sampling and randomization in Ecuador

2.3 Central African Republic (CAR)

Study in CAR was implemented in two Prefectures: Mbomou and Nana-Gribizi (he two most accessible areas (aside from Bangui) that also had a varying level of food security).

The selection of PSUs followed a similar pattern using field access (security situation) and mobile phone network coverage. The list of Admin5 areas that met the access and mobile phone coverage criteria formed the sampling frame. A total of 25 Admin5 areas (villages) were randomly selected from the list of Admin5 areas compiled jointly be the CAR CO and the local mobile phone service providers (to ensure that all Admin5 areas selected had mobile phone network). Admin5 areas without phone coverage were automatically excluded.

To ensure that the final sample was a mix of rural and urban households (taking into consideration there was a potentially different food security situation between rural and urban households), sub layer of locality was included during the selection of Admin5 areas so that the final sampled PSUs were a combination rural and urban areas.

The sampling frame for the study was all households in the 25 selected ADMIN5 areas. The listing phase was conducted through the random-walk methodology in each selected PSU by trained enumerators. A minimum of 100 households/respondents per PSU totalling 2500 households/respondents were listed as potential study participants. Phone numbers of the main respondent and any other phone numbers within the household that could have been used to contact the main respondent during the survey were collected. Drawing from the data collection experience in Myanmar and Ecuador, a number of demographic characteristics such as age of household head,

gender of household head, household size, and ownership of common household assets was collected during household listing and used for the statistical balance test.

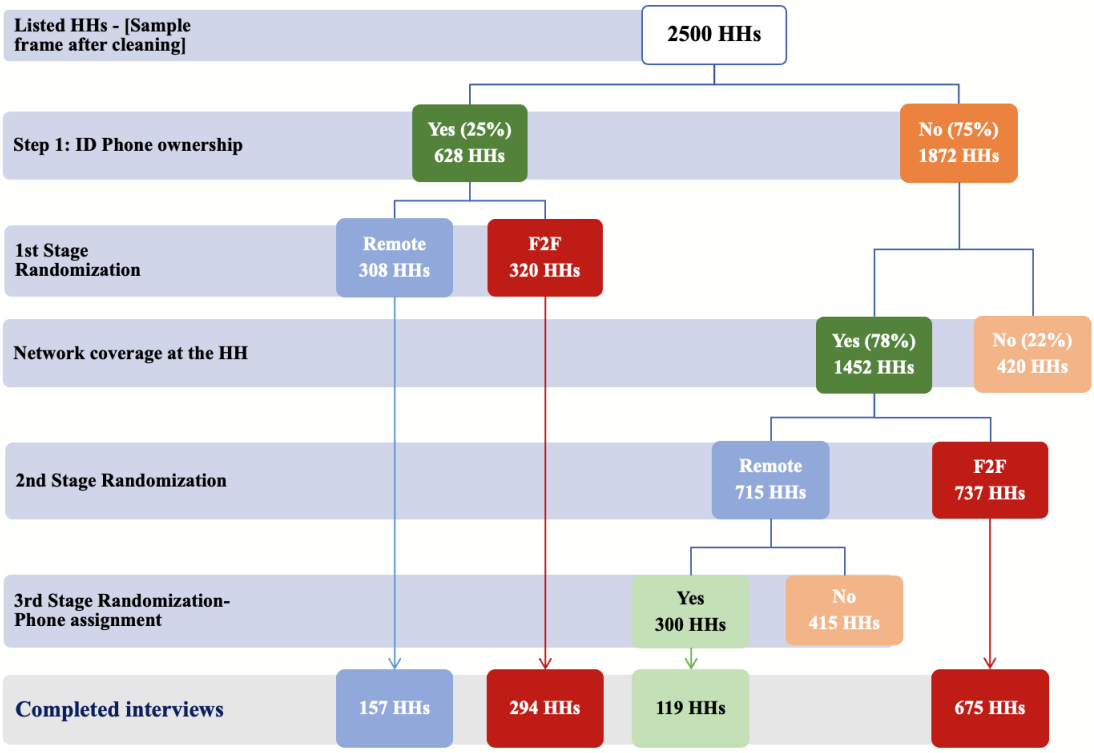

Figure 3: Sample Composition and Completion in CAR

2.4 Burkina Faso

Following in the study framework illustrated in Figure 1, the first three steps have already been completed. In conjunction with the Burkina Faso CO, two prefectures (ADMIN1s) namely: Centro and North prefectures have been selected to be the study areas (selection based on accessibility and phone coverage).

Based on the list of ADMIN5 areas available to the CO, an initial list of 25 villages (Admin 5 areas) have been randomly selected to form the main PSUs from which the participating households would be selected following a random walk procedure.

The household sampling frame will be made up of all households in each of the selected 25 PSUs. Households would be enlisted into the study through household listing process that would be implemented through a random walk by trained enumerators in each PSU. Starting from a known point, the enumerators will select every 3rd household in a systematic way.

Households enlisted but that do not have access to mobile phones or phone coverage will be dropped. Also, households with incomplete baseline and contact information will be dropped from the final

sample to be randomized.